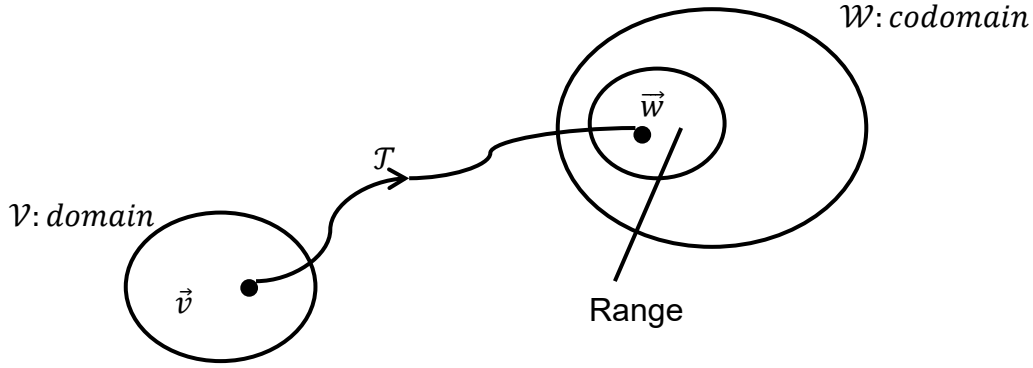
6.1: Introduction to Linear Transformations

MATH 2360

Definitions and Theorems:

1. A function that maps a vector space $\mathcal{V}$ onto a vector space $\mathcal{W}$ is denoted $\mathcal{T}: \mathcal{V} \to \mathcal{W}$, where $\mathcal{V}$ is the **domain** and $\mathcal{W}$ is the **codomain**. If $\vec{v}$ is in $\mathcal{V}$ and $\vec{w}$ is in $\mathcal{W}$ such that $\mathcal{T}(\vec{v}) = \vec{w}$, then $\vec{w}$ is the **image** of $\vec{v}$ under $\mathcal{T}$. The set of all such images is the *range* of $\mathcal{T}$. The set of all vectors $\vec{v}$ in $\mathcal{V}$ such that $\mathcal{T}(\vec{v}) = \vec{w}$ is the **preimage** of $\vec{w}$.



Examples

1) For any vector $\vec{v} = (v_1, v_2) \in \mathbb{R}^2$, define $\mathcal{T}: \mathbb{R}^2 \to \mathbb{R}^2$ by $\mathcal{T}(v_1, v_2) = (v_1 + v_2, v_1 - v_2)$. Find
   a) Image $\vec{v} = (3, -4)$

   b) Preimage of $\vec{w} = (3, 19)$

Definitions

1. **Linear Transformation:** Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces. The function $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ is a **linear transformation** of $\mathcal{V}$ into $\mathcal{W}$ when, for all $\vec{u}, \vec{v} \in \mathcal{V}$ and for any scalar $c \in \mathbb{R}$,
   (1) $\mathcal{T}(\vec{u} + \vec{v}) = \mathcal{T}(\vec{u}) + \mathcal{T}(\vec{v})$ and
   (2) $\mathcal{T}(c\vec{u}) = c\mathcal{T}(\vec{u})$

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Examples

2) Show that the function in the previous example is a linear transformation.

$$\mathcal{T}: \mathbb{R}^2 \to \mathbb{R}^2 \text{ by } \mathcal{T}(v_1, v_2) = (v_1 + v_2, v_1 - v_2)$$

2. ***Note:*** Examples of functions that are ***not*** linear transformations include
   a) $f(x) = cos(x)$
   b) $f(x) = x^3$
   c) $f(x) = x - 5$

3. **Properties of Linear Transformations:** If $\mathcal{T}$ is a linear transformation from $\mathcal{V}$ onto $\mathcal{W}$, where $\vec{u}, \vec{v} \in \mathcal{V}$, then:
   (1) $\mathcal{T}(\vec{0}) = \vec{0}$ (the zero transformation)
   (2) $\mathcal{T}(-\vec{v}) = -T(\vec{v})$
   (3) $\mathcal{T}(\vec{u} - \vec{v}) = \mathcal{T}(\vec{u}) - \mathcal{T}(\vec{v})$
   (4) $\mathcal{T}(\vec{v}) = \mathcal{T}(c_1 v_1 + c_2 v_2 + \cdots + c_n v_n) = c_1 \mathcal{T}(v_1) + c_2 \mathcal{T}(v_2) + \cdots + c_n \mathcal{T}(v_n)$
   (5) ***Note:*** $\mathcal{T}(\vec{v}) = \vec{v}$ is the identity transformation.

Examples

3) Let $\mathcal{T}: \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that $\mathcal{T}(1, 0, 0) = (2, 4, -1)$, $\mathcal{T}(0, 1, 0) = (1, 3, -2)$, and $\mathcal{T}(0, 0, 1) = (0, -2, 2)$. Find $\mathcal{T}(1, -3, 0)$.

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Definitions

4. **Matrix Transformation:** Let A be an $m \times n$ matrix. The function $\mathcal{T}$ defined by $\mathcal{T}(\vec{v}) = A\vec{v}$ is a $\mathcal{T}(\vec{v}) = A\vec{v}$ is a linear transformation from $\mathbb{R}^n$into $\mathbb{R}^m$ ($n \times 1$ matrices represent vectors in $\mathbb{R}^n$, $m \times 1$ matrices represent vectors in $\mathbb{R}^m$).

   Note: A 2×3 matrix ⇒ transformation from $\mathbb{R}^3 \to \mathbb{R}^2$, a 4×2 matrix ⇒ transformation from $\mathbb{R}^2$ into $\mathbb{R}^4$.

Examples

4) Show that $\mathcal{T}: \mathbb{R}^2 \to \mathbb{R}^2$ represented by $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ has the property that it rotates every vector in $\mathbb{R}^2$counterclockwise about the origin through θ.

5. The linear transformation is a **rotation** in $\mathbb{R}^2$. (A rotation preserves vector length ($\|\mathcal{T}(\vec{u})\| = \|\vec{u}\| \text{ and } \|\mathcal{T}(\vec{v})\| = \|\vec{v}\|$) and preserves the angle between two vectors (the angle between $\mathcal{T}(\vec{u})$ and $\mathcal{T}(\vec{v})$ is equal to the angle between $\vec{u}$ and $\vec{v}$).)

6. The linear transformation $\mathcal{T}: \mathbb{R}^3 \to \mathbb{R}^3$ represented by $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is called a **projection** in $\mathbb{R}^3$. It maps every vector in $\mathbb{R}^3$ to its orthogonal projection in the xy-plane.

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